

Prakshit Mathur

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SUMMARY

I am a passionate ML enthusiast and 3rd-year CS undergraduate (CGPA 8.55/10, BIT Mesra) who genuinely loves exploring how machines learn and make decisions. Driven by curiosity, I built end-to-end AI/ML projects across NLP, LLMs, RAG, and generative deep learning. Oracle Data Science certified and I am eager to bring hands-on energy to a team where I can learn fast, contribute faster, and grow.

PROJECTS

- RAG-Based Intelligent Document Assistant** Python · LLMs · Embeddings · FAISS · FastAPI [Live](#) 2026
 - Developed a RAG-powered document assistant that enables users to ask questions over PDFs **using semantic search** and LLM-based response generation.
 - Integrated FAISS vector search and embedding-based retrieval to provide **context-aware answers** while minimizing irrelevant responses.
- AI Research Paper Analyzer** Python · RAG · LLM · FastAPI [Github](#) 2026
 - Built an AI-powered research assistant that aggregates papers from multiple academic sources and ranks them based on relevance.
 - Added semantic search and **intelligent ranking features**, allowing researchers to explore papers using natural language instead of manual keyword searches.
- Text-to-Music Generation using Transformers** Python · PyTorch · TransformerArchitecture [Live](#) 2024
 - Developed a deep learning application that converts text prompts into music using **Transformer-based architectures** and PyTorch.
 - Experimented with deep learning techniques for **sequence generation** and audio synthesis using PyTorch.
- SMS Spam Detection – Production NLP** Python · TF-IDF · MultinomialNaive Bayes · FastAPI [Live](#) 2025
 - Built an NLP-based spam detection system using TF-IDF and **Multinomial Naive Bayes**, achieving 92% classification accuracy on SMS datasets.
 - Deployed the model through a FastAPI service with automated preprocessing and confidence scoring for real-time predictions.

EXPERIENCE

- Machine Learning Trainee** 2025
 - Built a cancer-prediction classification pipeline achieving 88% CV score — executed the full ML lifecycle: data cleaning, EDA, feature engineering, **hyperparameter tuning**, and evaluation reporting.
- Python & ML Intern — YBI Foundation** 2024
 - Applied regression and classification algorithms on structured datasets; delivered documented model comparison reports covering accuracy, **precision, recall, and F1 score**.

TECHNICAL SKILLS

- Languages** : Python, C++
- AI / ML** : NLP, LLMs, RAG, transformer-based models, Neural Networks, classical ML
- Libraries** : PyTorch, Scikit-learn, LangChain, Pandas, NumPy, FastAPI, OpenCV, Matplotlib, Github, Streamlit
- Concepts** : Multi-Agents, Vector Stores, Embeddings, Semantic Search, DSA, Data Preprocessing

EDUCATION

Birla Institute of Technology, Mesra

Bachelor of Technology – Computer Science and Engineering | CGPA: 8.51

2023 –Present

CERTIFICATIONS

- Oracle Data Science Professional Certified** – validated end-to-end ML lifecycle including model building, evaluation & deployment.
- 150+ LeetCode** problems solved — demonstrates strong algorithmic problem-solving under timed competitive conditions.
- 5-Star HackerRank rating in C++** — in the top percentile for language proficiency, active rated competitor on CodeChef and Codeforces.